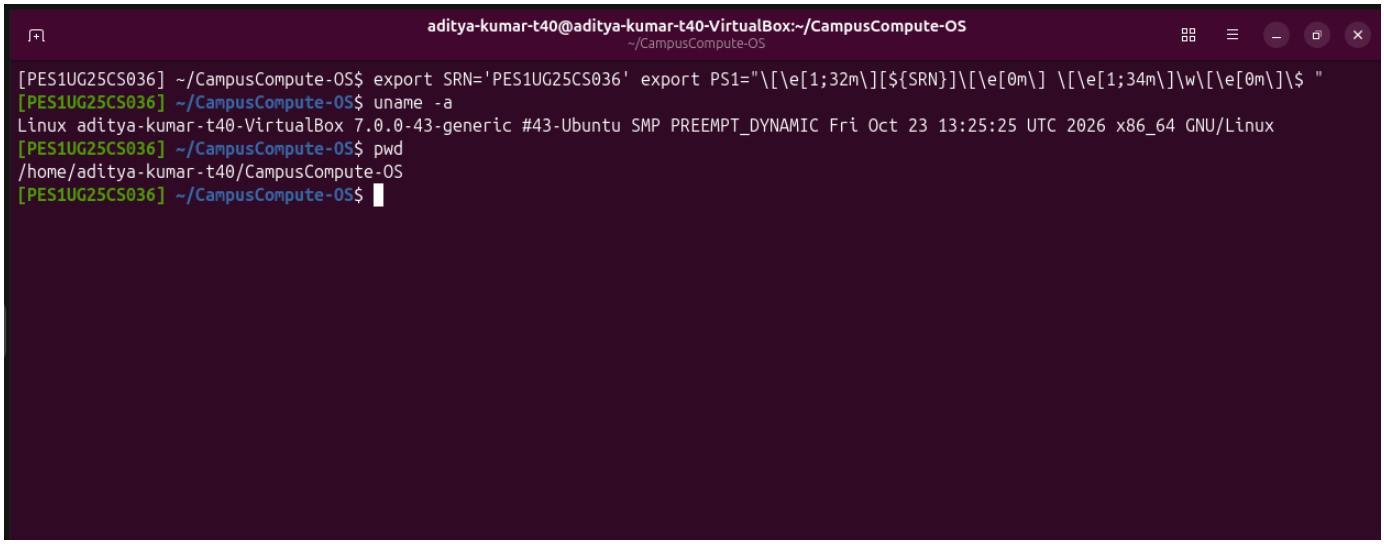
CampusCompute: Intelligent Job Scheduling & Resource Management

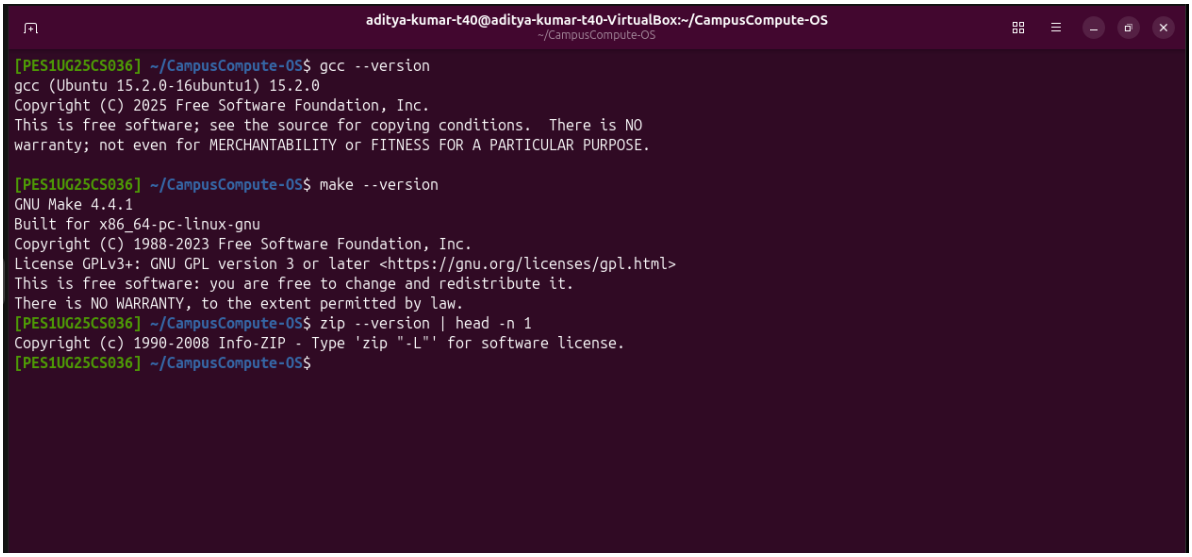
TEAM MEMBER 1: ADITYA KUMAR P | SRN: PES1UG25CS036
TEAM MEMBER 2: AKSHAY SWAMY S B | SRN: PES1UG25CS047
Operating Systems (UE25CS242A) — Level 3 Mini-Project | Screenshots S01–S24

S01 – Environment and SRN prompt



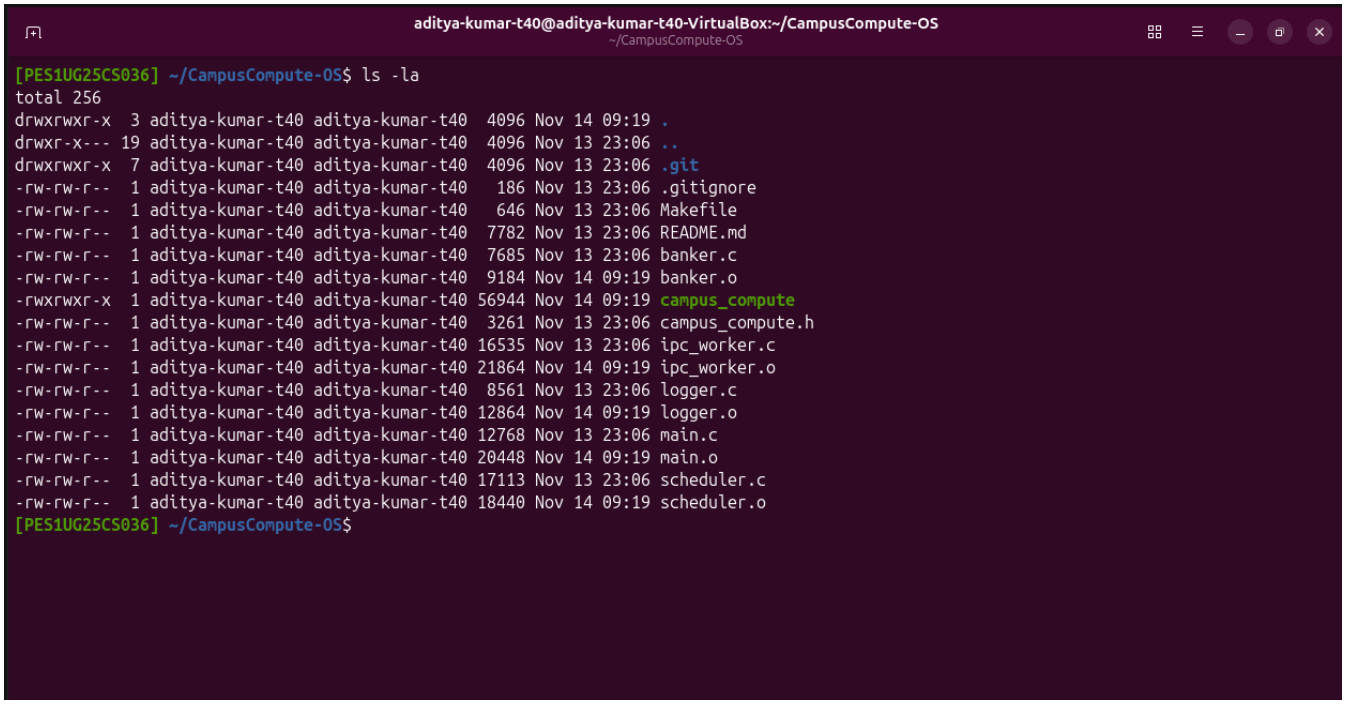
S01 – Shows the Ubuntu Linux terminal environment, the student's complete SRN (PES1UG25CS036) configured in the shell prompt, and the CampusCompute working directory.

S02 – Compiler and build tools



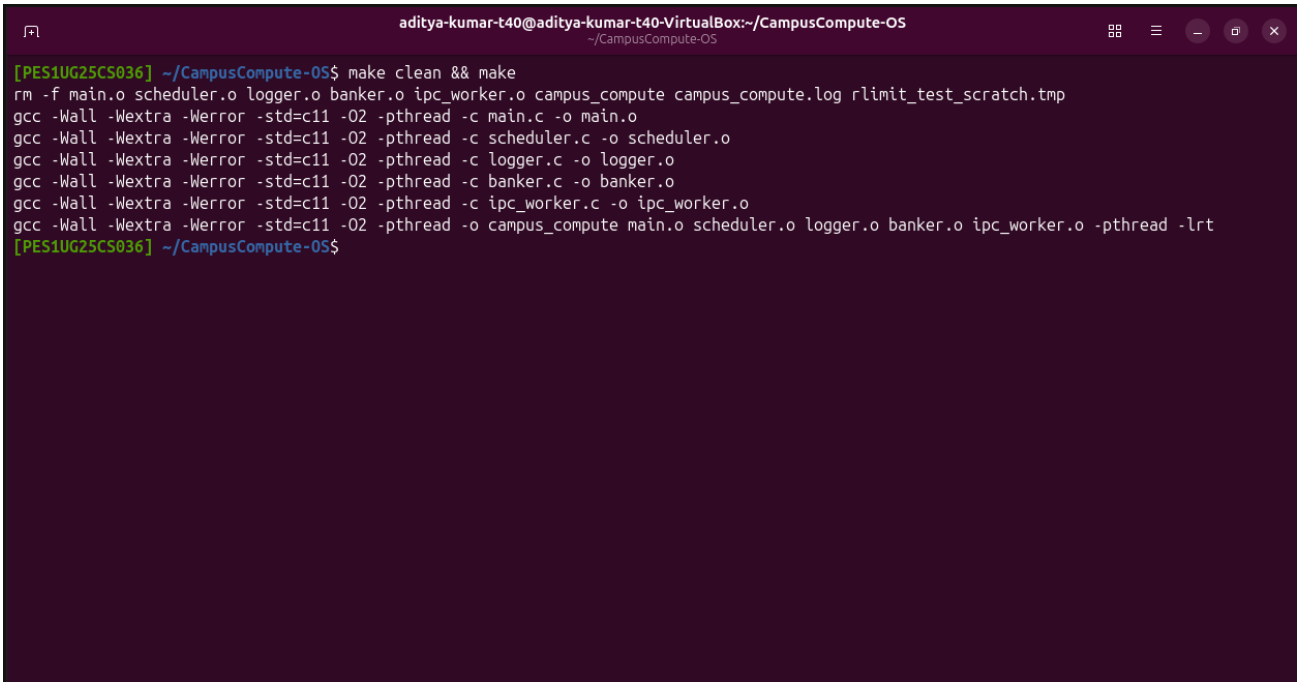
S02 – Confirms that GCC, GNU Make, and required build tools are installed and available in the Ubuntu environment.

S03 – Project files



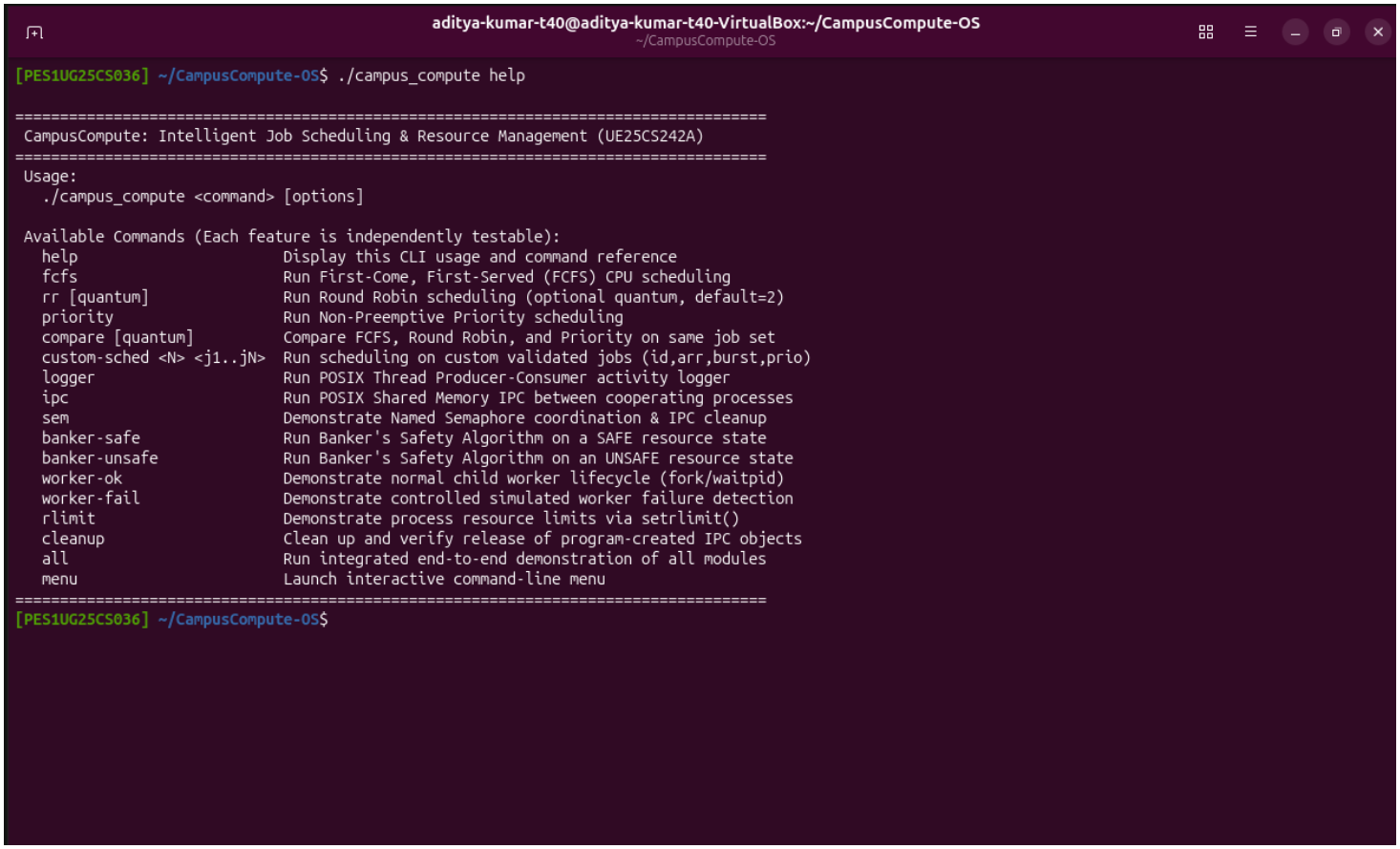
S03 – Displays the CampusCompute project directory containing all modular C source files, header file, Makefile, and README.

S04 – Clean build



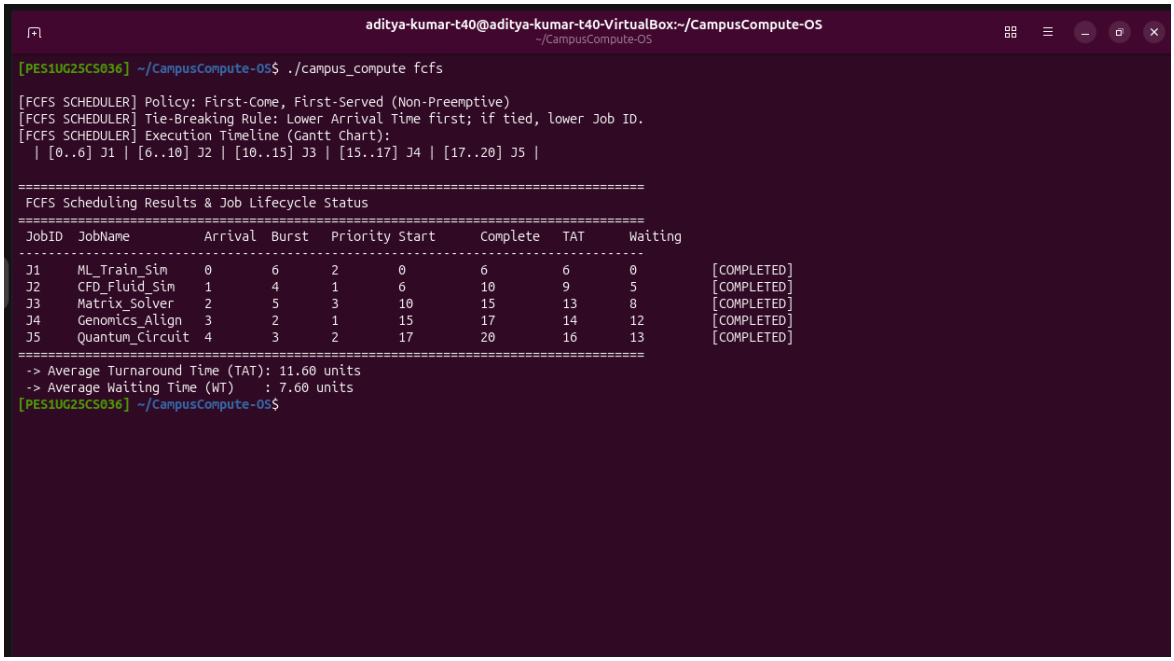
S04 – Demonstrates the CampusCompute program compiling cleanly from source using Make with zero warnings or compiler errors.

S05 – CLI help/usage



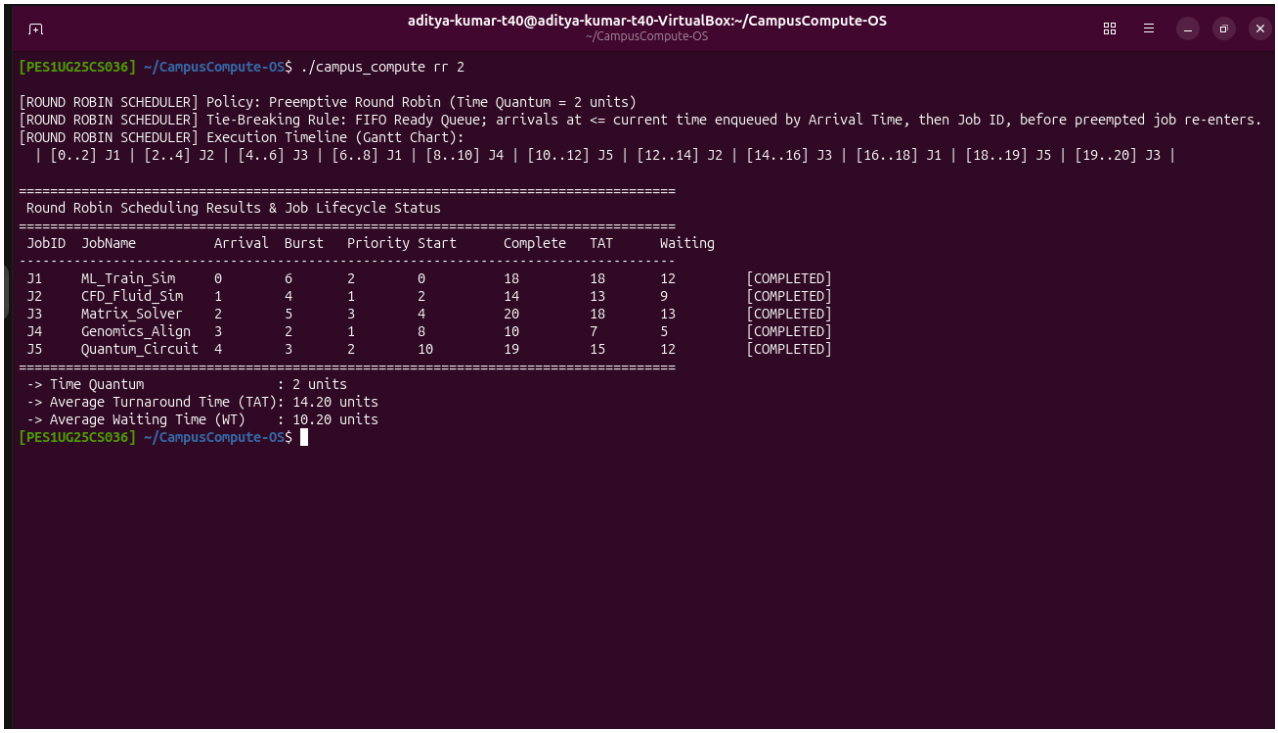
S05 – Shows the command-line interface help output listing all available commands, syntax, and interactive menu options.

S06 – FCFS scheduling



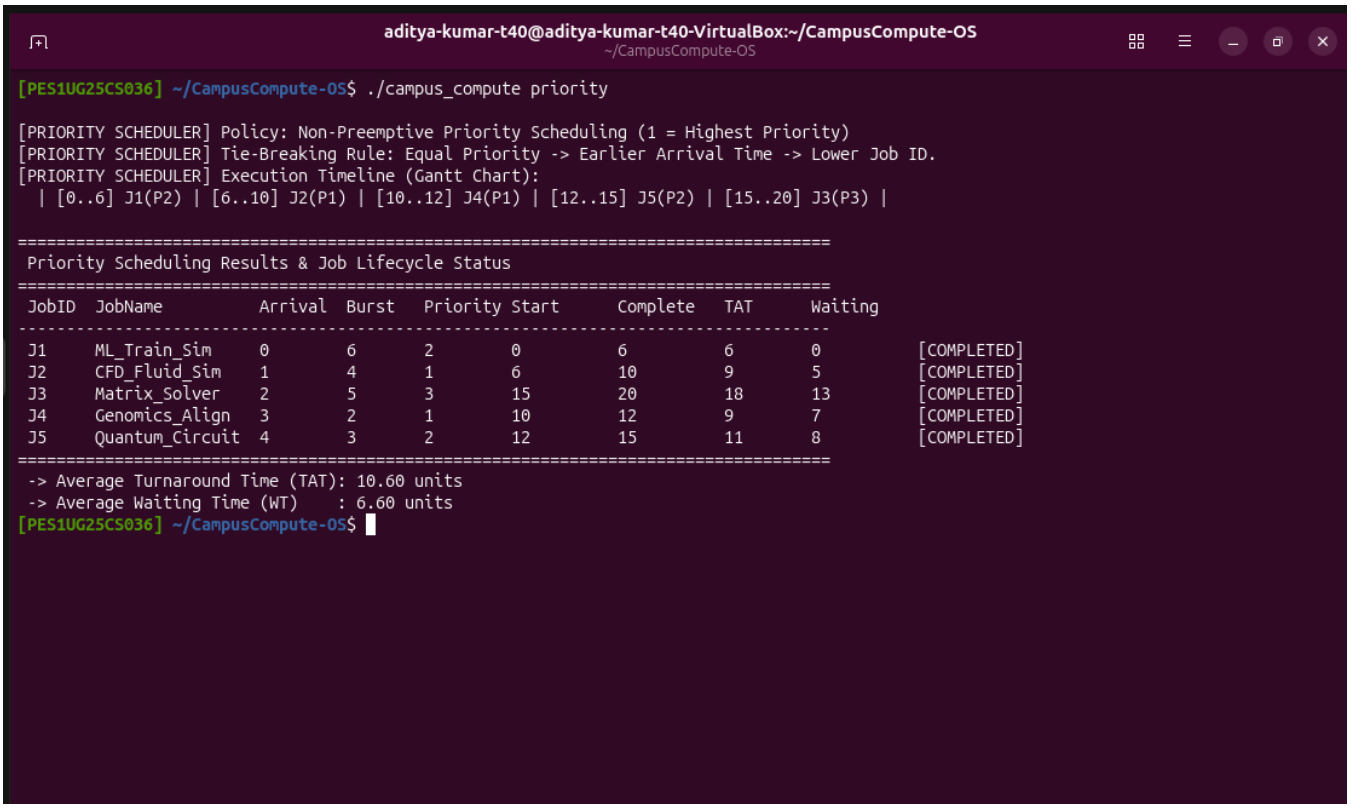
S06 – Demonstrates First-Come, First-Served (FCFS) CPU scheduling with its execution order timeline, tie-breaking rule, and calculated turnaround and waiting times.

S07 – Round Robin scheduling



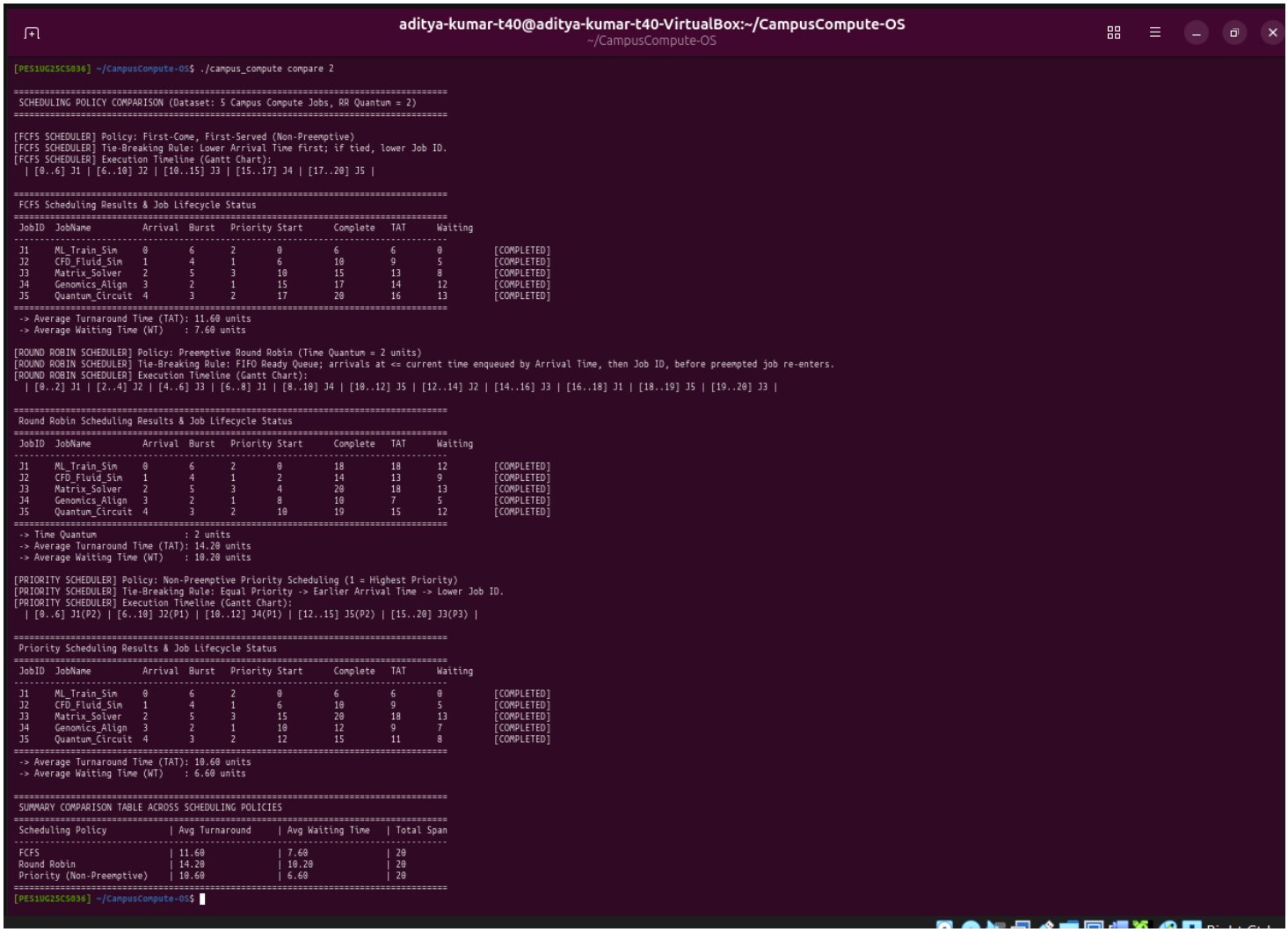
S07 – Demonstrates preemptive Round Robin scheduling with a time quantum of 2 units, Gantt execution timeline, and per-job scheduling metrics.

S08 – Priority scheduling



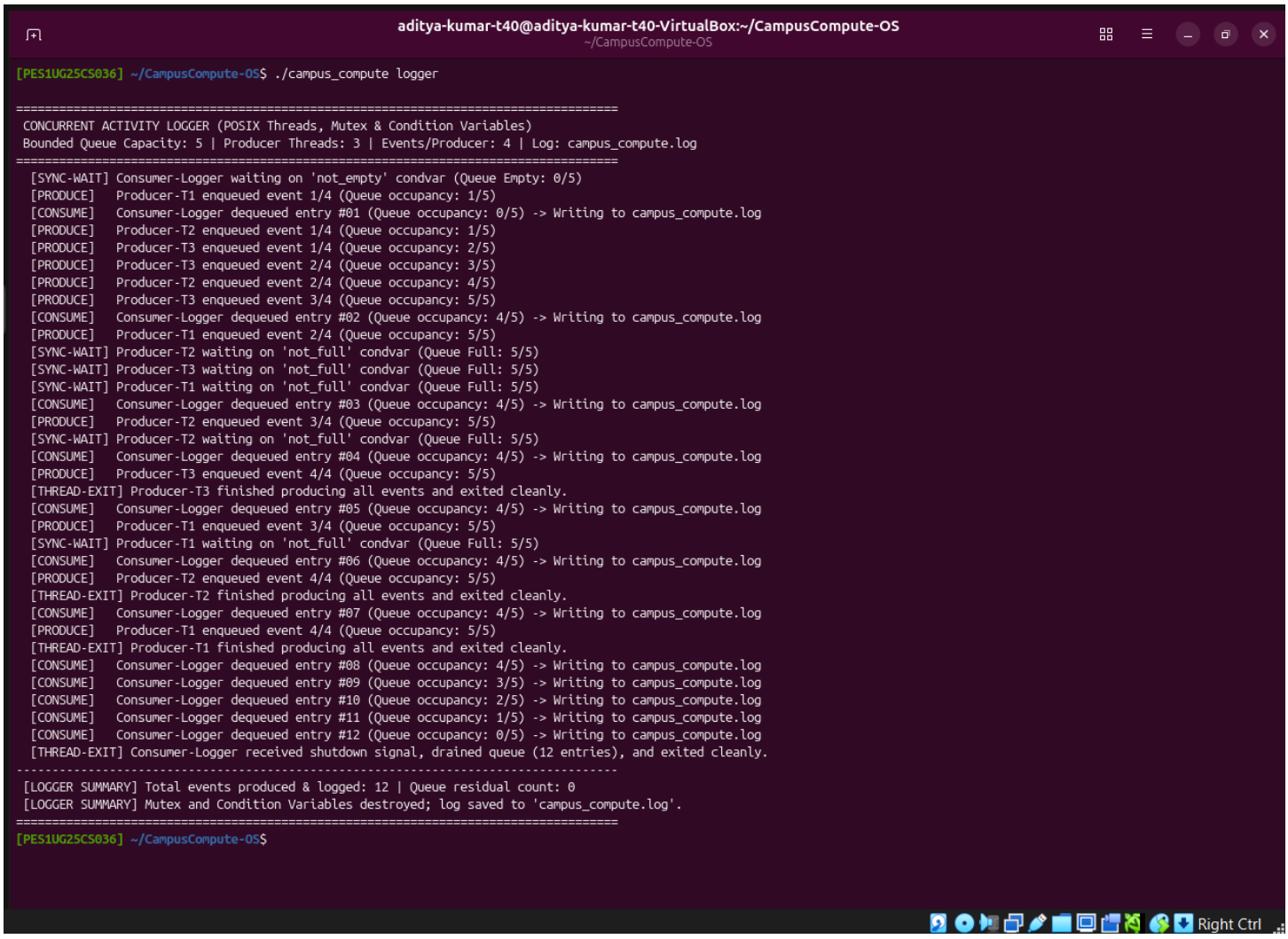
S08 – Shows non-preemptive Priority Scheduling (where lower number indicates higher priority), tie-breaking rule, and calculated scheduling metrics.

S09 – Scheduling comparison



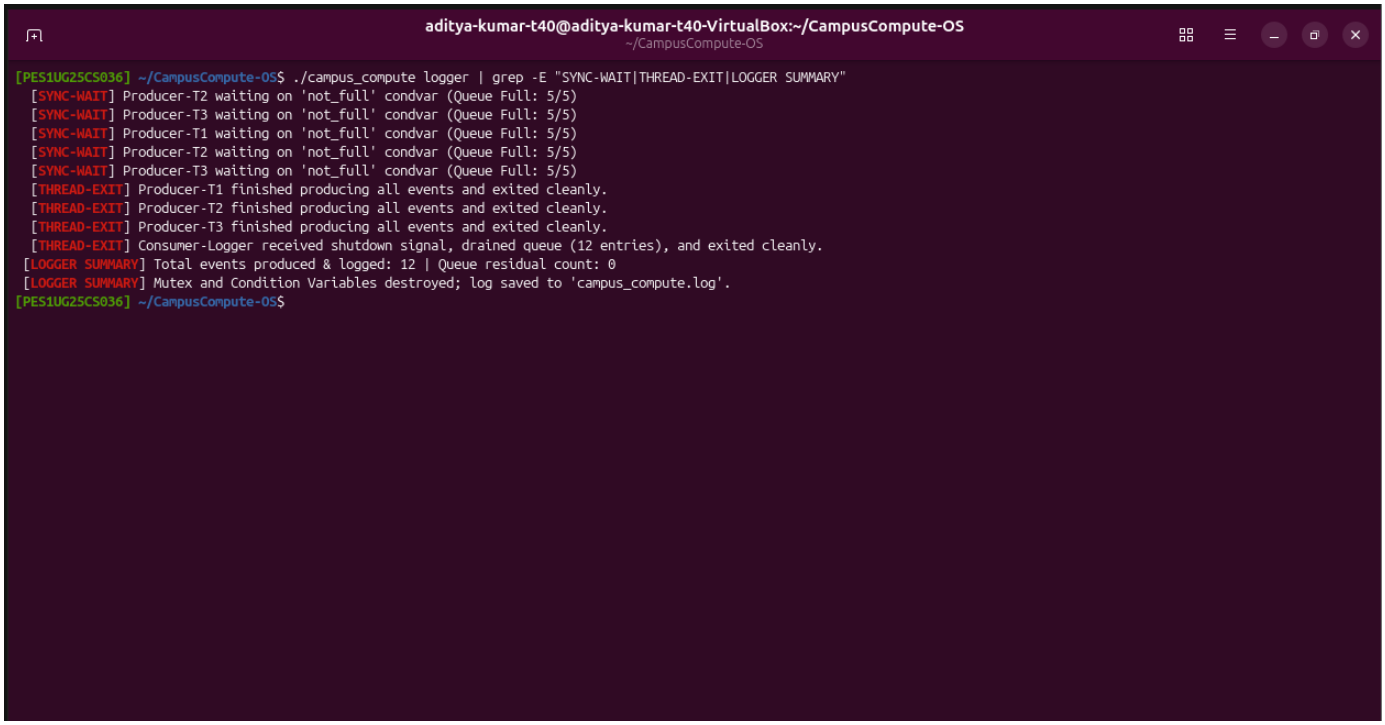
S09 – Compares FCFS, Round Robin (Quantum = 2), and Non-Preemptive Priority scheduling side-by-side on the identical 5-job workload.

S10 – Producer-consumer logger



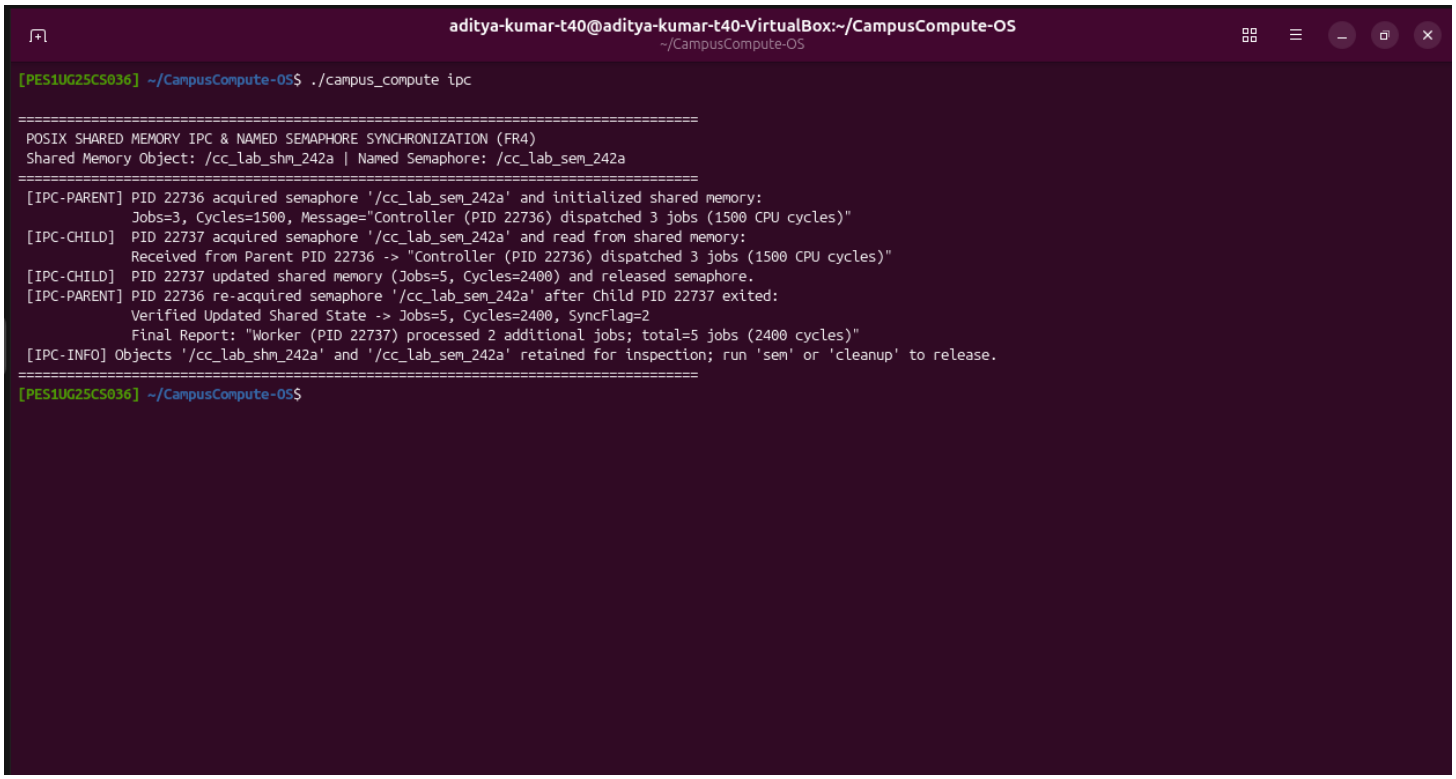
S10 – Shows concurrent producer threads enqueuing job events and the consumer logger thread dequeuing entries from the bounded queue (capacity 5) into campus_compute.log.

S11 – Thread synchronization and termination



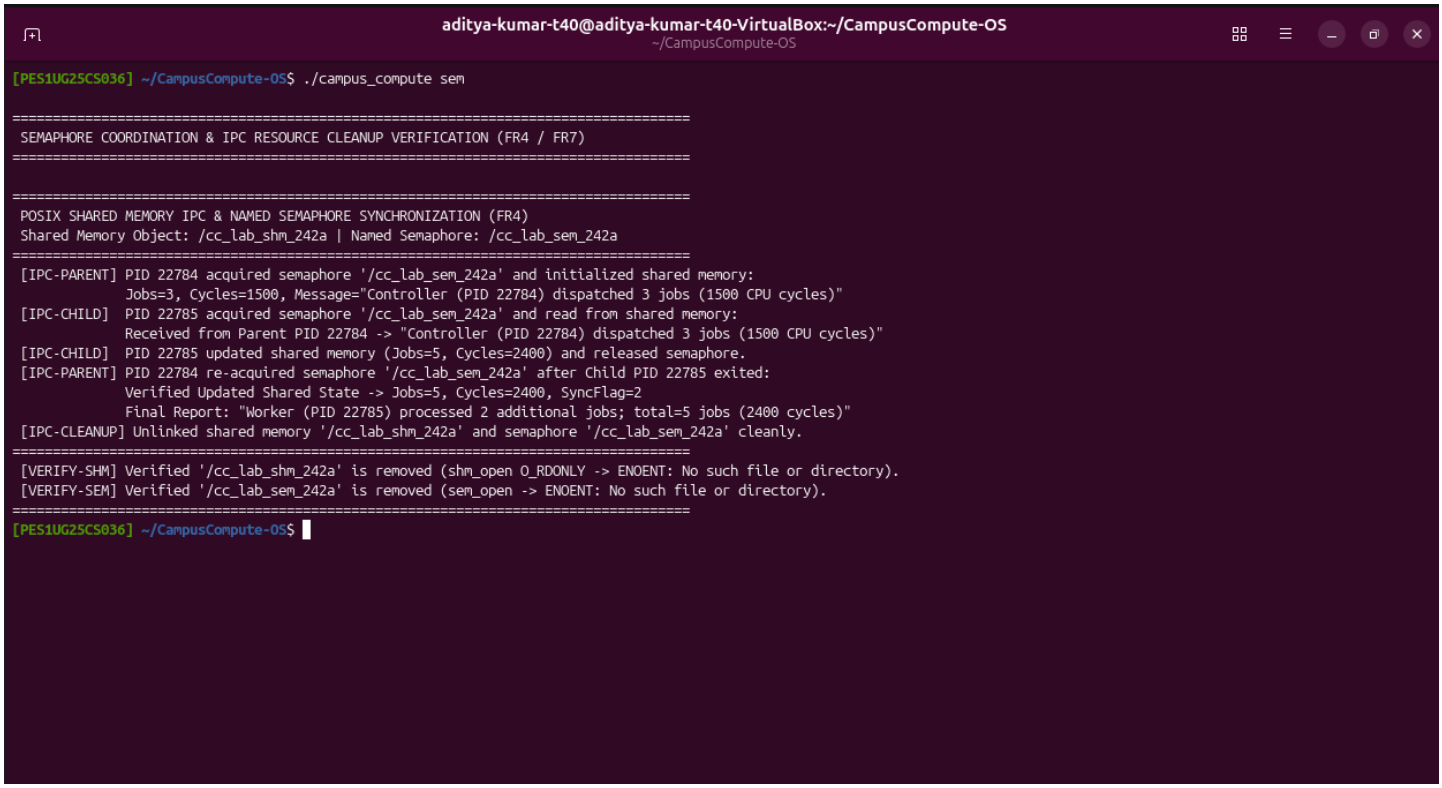
S11 – Demonstrates condition-variable synchronization (waiting on full/empty queue states) and orderly termination of all producer and consumer threads.

S12 – Shared-memory IPC



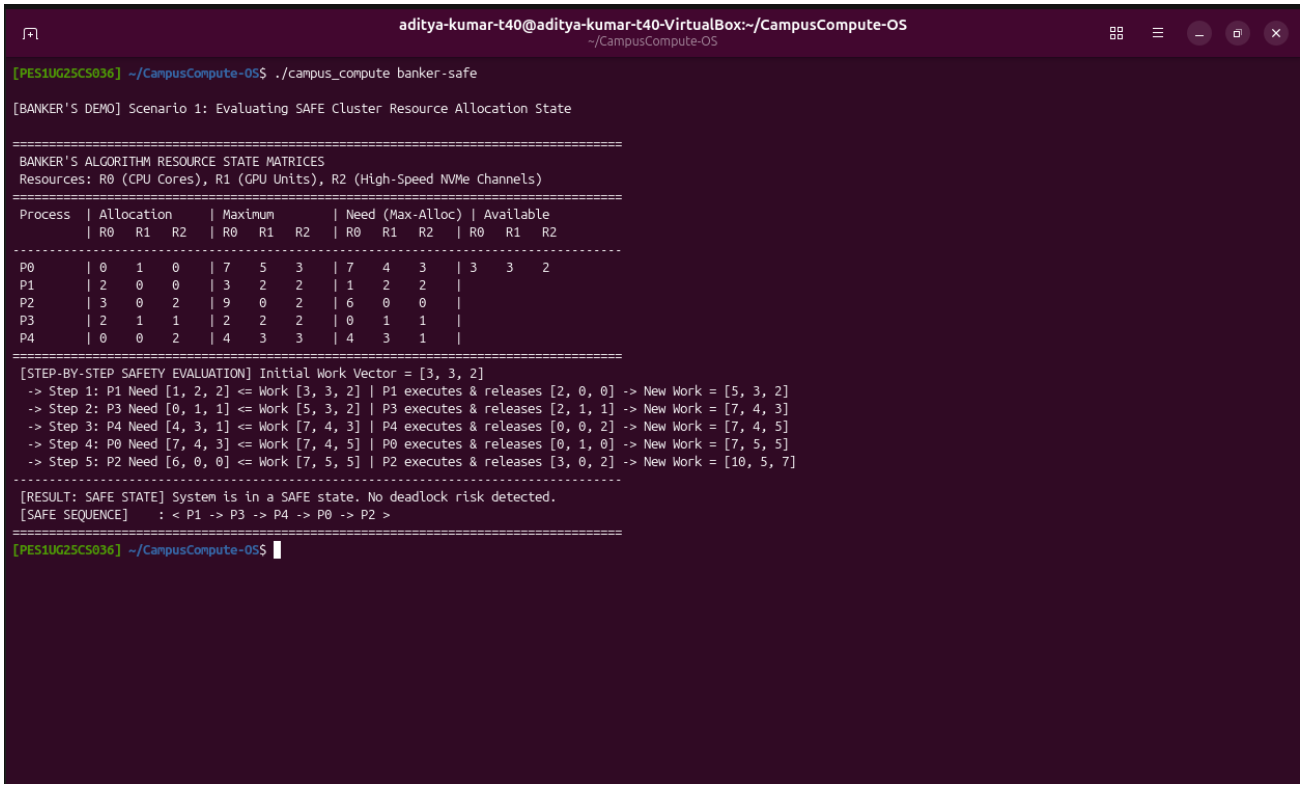
S12 – Demonstrates cooperating parent and child processes exchanging cluster telemetry through POSIX shared memory (/cc_lab_shm_242a).

S13 – Semaphore coordination and cleanup



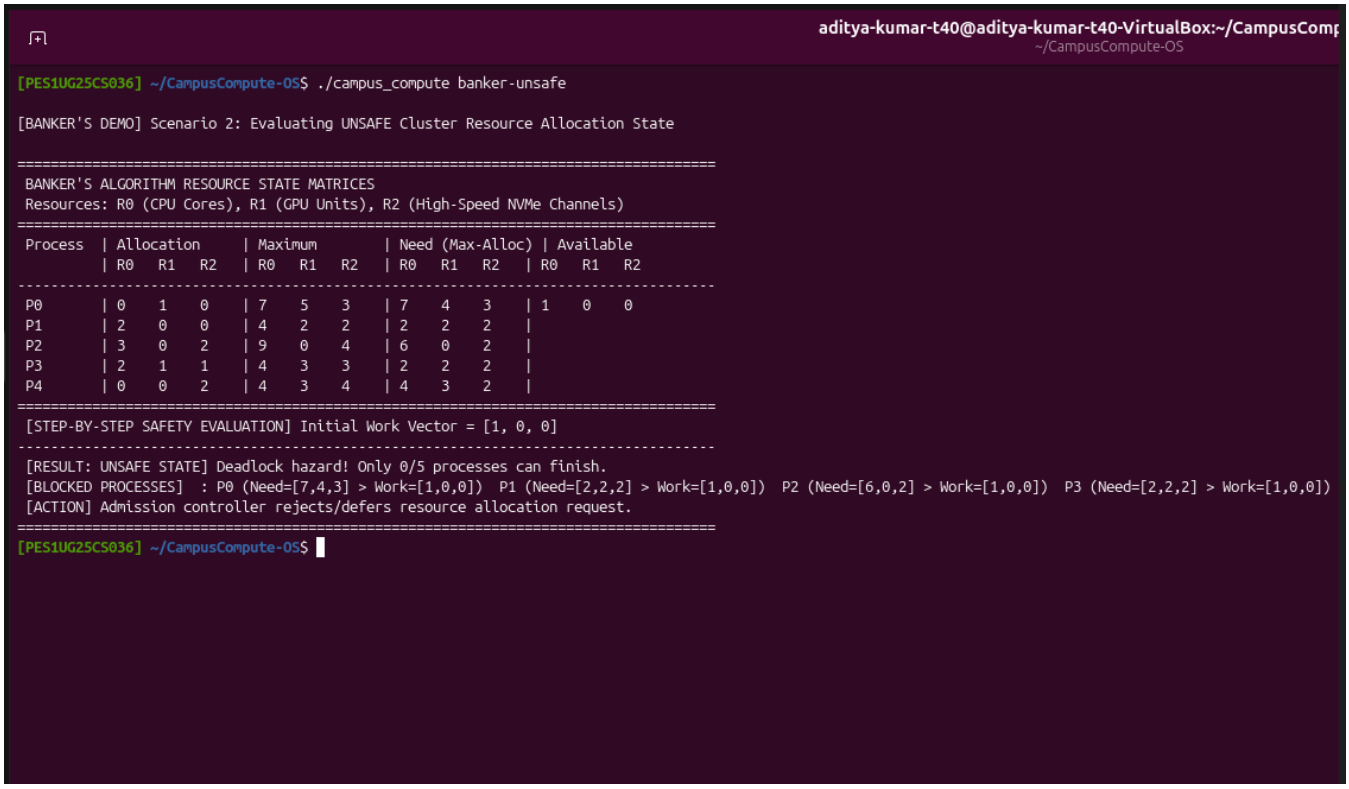
S13 – Shows mutual exclusion using the POSIX named semaphore (/cc_lab_sem_242a) and verifies complete unlinking and cleanup of created IPC resources.

S14 – Banker's safe state



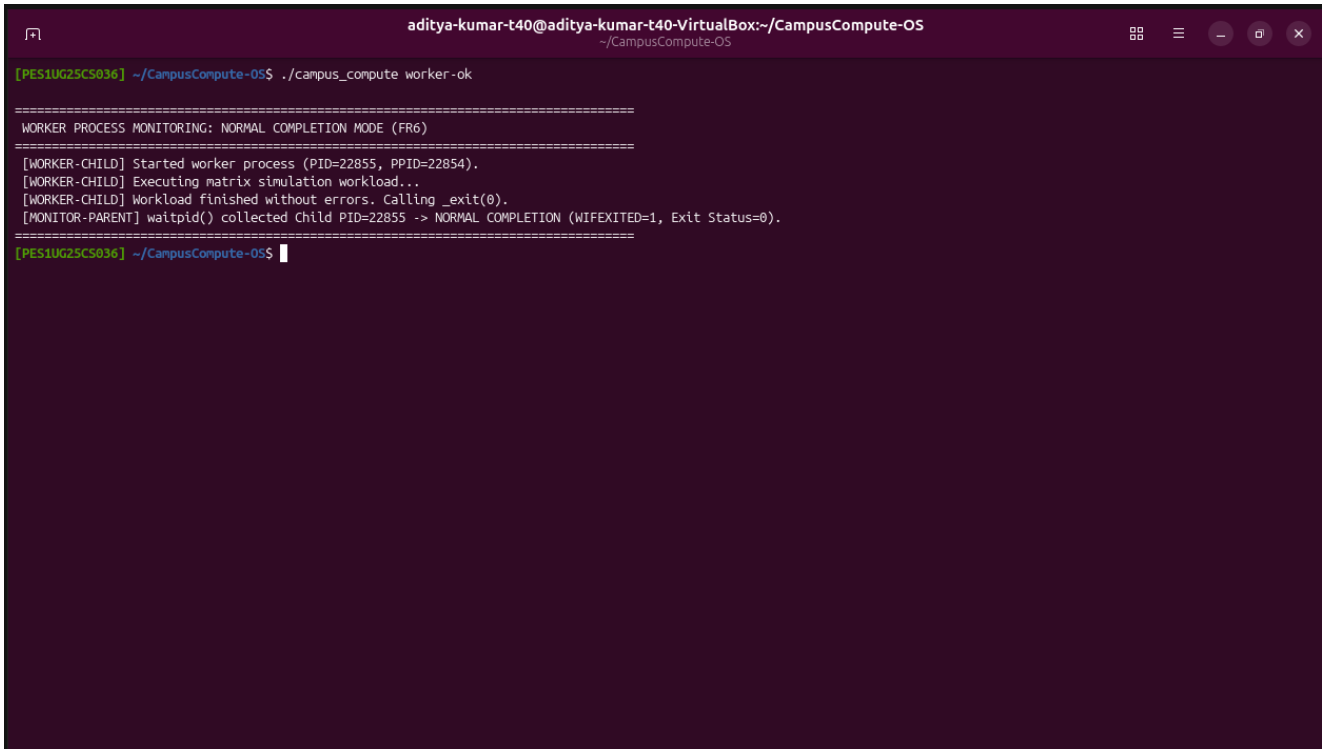
S14 – Demonstrates the Banker's safety algorithm evaluating a safe resource state and producing the valid safe execution sequence <P1 -> P3 -> P4 -> P0 -> P2>.

S15 – Banker's unsafe state



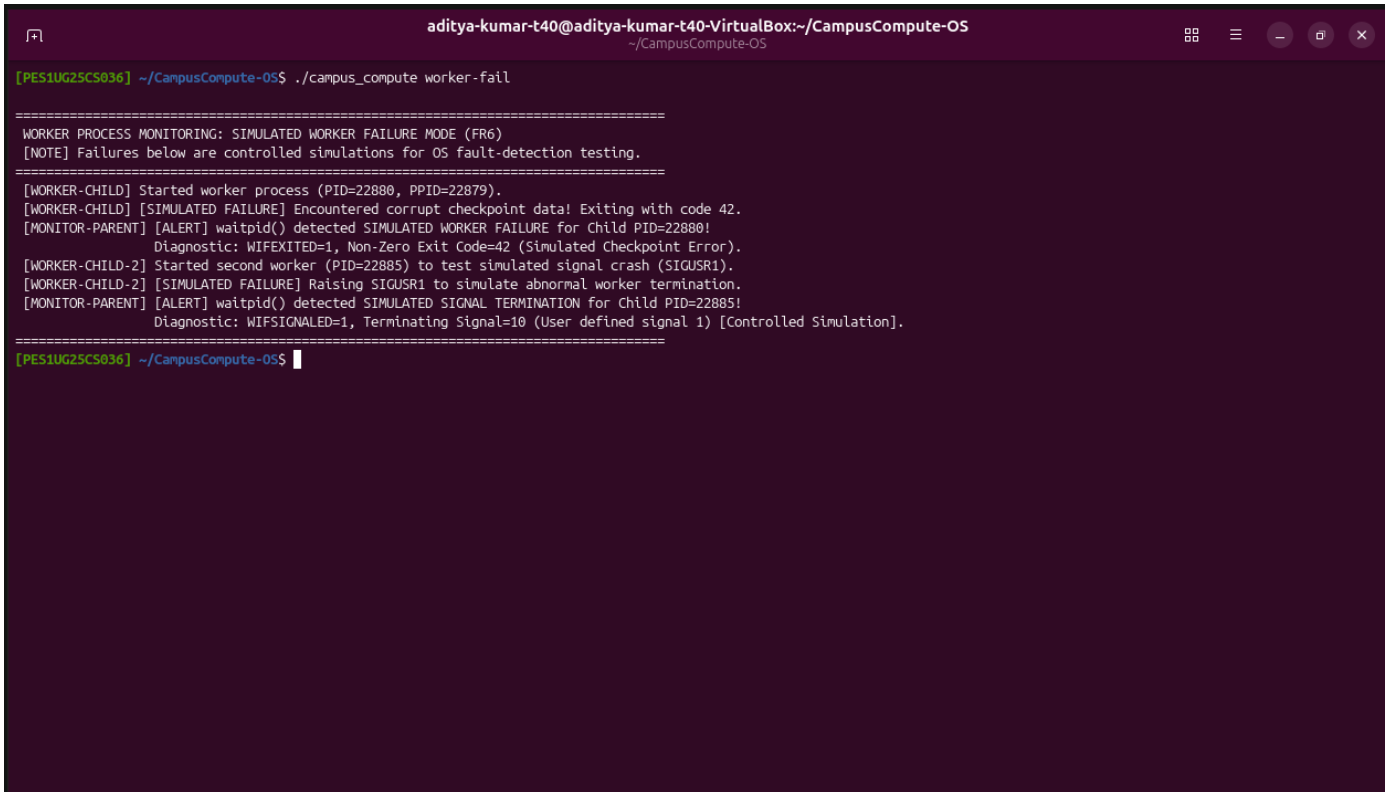
S15 – Shows the Banker's safety algorithm detecting an unsafe resource state where remaining process needs exceed available work, preventing deadlock.

S16 – Normal worker process



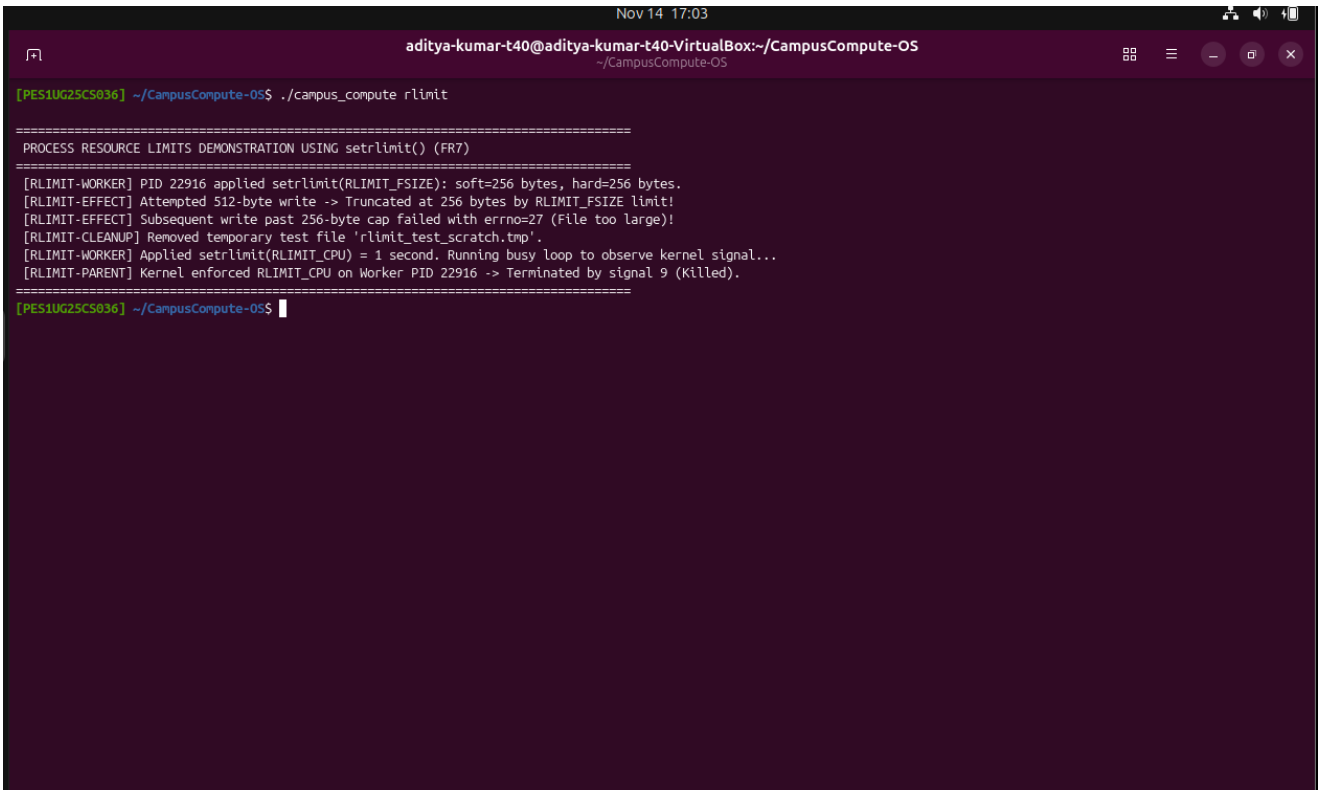
S16 – Shows child worker process creation via fork() and parent collection of normal completion status (exit status 0) via waitpid().

S17 – Simulated worker failure



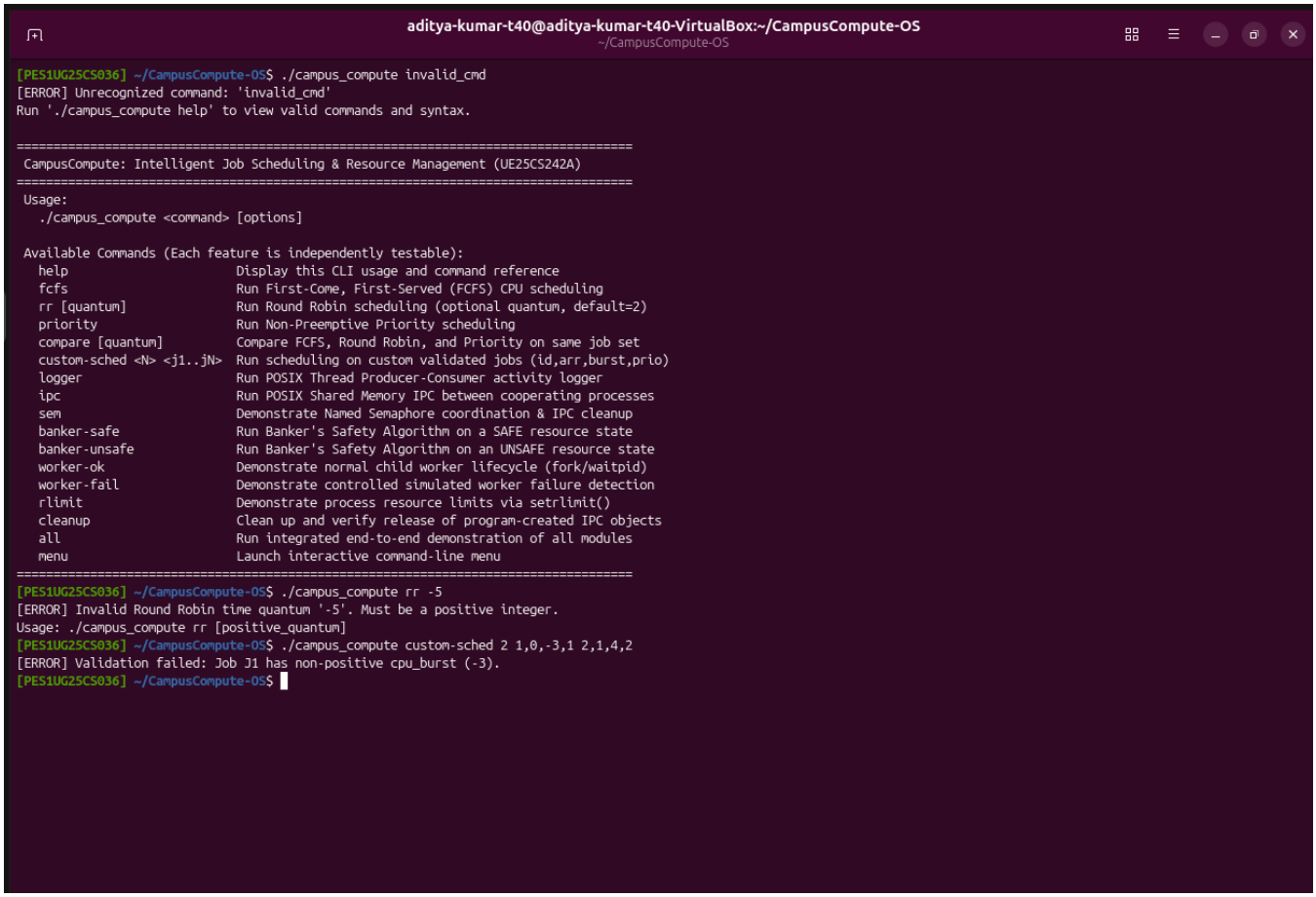
S17 – Demonstrates controlled simulated worker failures (non-zero exit code 42 and SIGUSR1 termination) and accurate parent-side failure detection.

S18 – Resource-limit demonstration



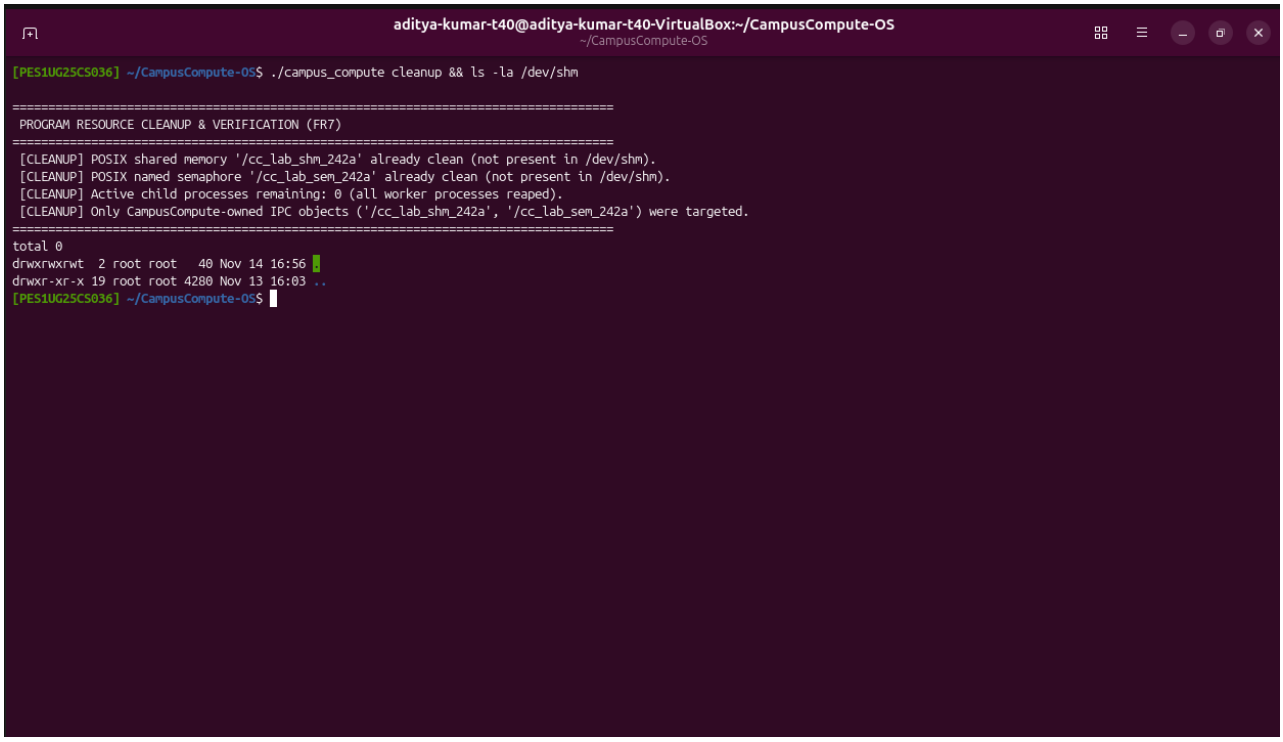
S18 – Demonstrates applying process resource limits via setrlimit() (RLIMIT_FSIZE = 256B and RLIMIT_CPU = 1s) and shows the observed kernel enforcement.

S19 – Invalid input/command



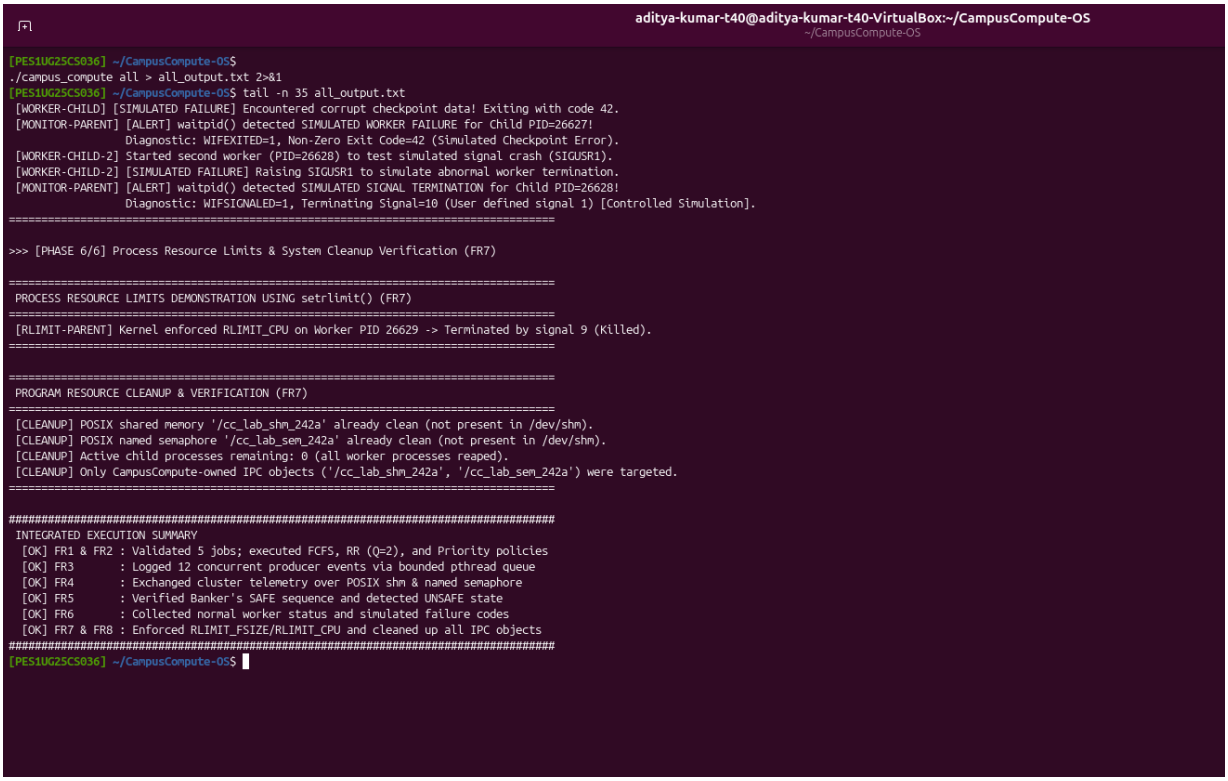
S19 – Shows error handling and usage messages when an unrecognized command, negative time quantum, or invalid job burst is provided.

S20 – Cleanup verification



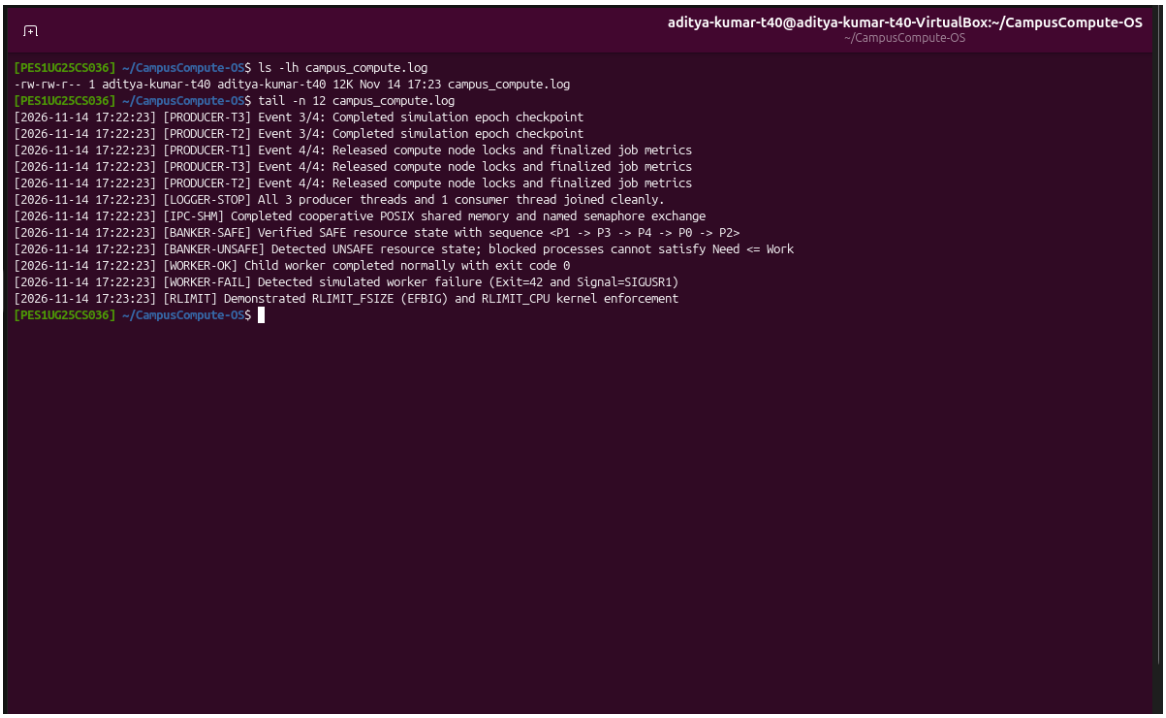
S20 – Confirms that all worker processes have ended and all CampusCompute-created shared memory and semaphore objects are released from /dev/shm.

S21 – Integrated execution



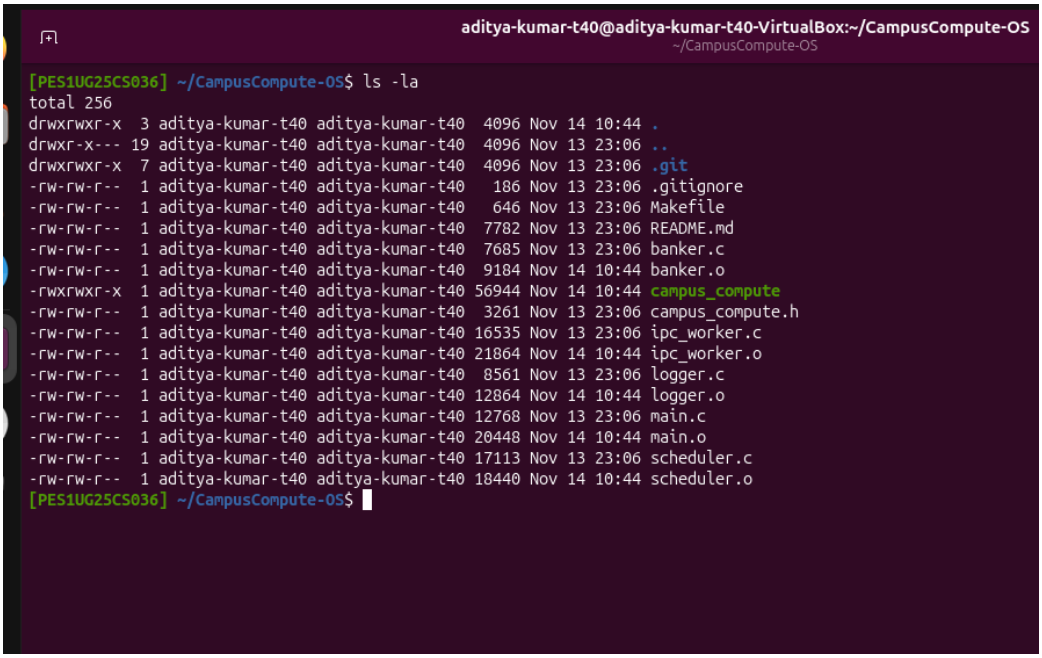
S21 – Shows the end-to-end integrated execution across all six CampusCompute subsystems and prints the final verification summary.

S22 – Generated logs



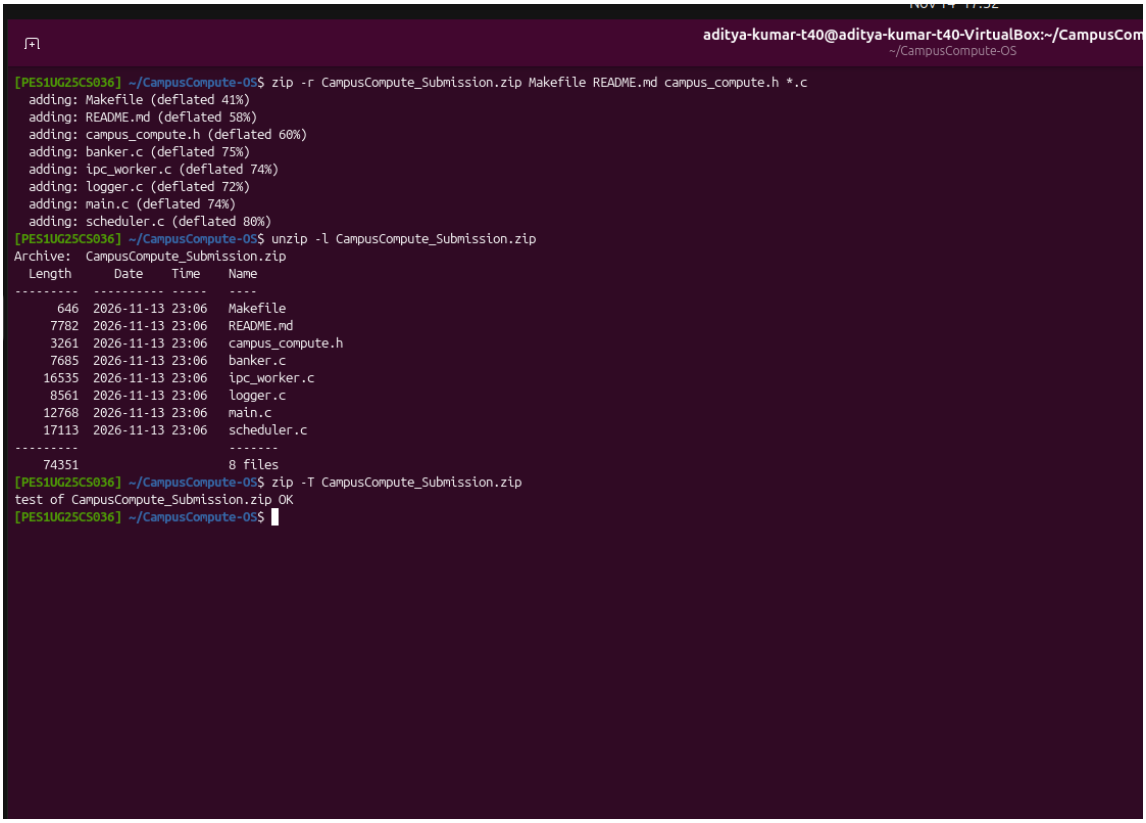
S22 – Displays the campus_compute.log file generated by the program along with a sample of its timestamped producer-consumer and subsystem log entries.

S23 – Final source and build files



S23 – Shows the final source files, header, Makefile, README, and compiled campus_compute executable in the project directory.

S24 – Submission ZIP



S24 – Shows the contents of the CampusCompute_Submission.zip archive and confirms a successful archive integrity check.